Problem Statement No. 22
Interview Trainer Agent (RAG Based)

The Challenge

Cracking a job interview takes more than technical knowledge. Candidates struggle to know which questions to expect for a given role, how to structure strong answers, and where their weaknesses lie. Interview preparation material is scattered across recruitment portals, blogs, and forums, and much of it is generic, outdated, or unverified. Human mock interviews and coaching are expensive and hard to scale, and they rarely adapt to the candidate's profile, experience level, or target role. As a result, many capable candidates underperform in competitive hiring environments.

The Objective

Build an AI-powered Interview Trainer Agent that prepares individuals for job interviews with personalized questions, model answers, and actionable feedback. The solution should:

- **Personalized Interview Preparation** – Generate tailored question sets and preparation strategies based on the candidate's resume, experience level, and target job role.
- **Interview Data RAG Retrieval** – Retrieve and summarize role-specific questions, industry expectations, behavioral scenarios, and HR guidelines from reliable sources such as recruitment portals, professional networks, and company interview databases.
- **Mock Interview & Feedback** – Conduct interactive mock interviews via text (and optionally voice), evaluate each answer for relevance, clarity, depth, and confidence, and give instant feedback with improvement tips and model answers.
- **Technical and Soft Skill Assessment** – Cover both domain-specific technical questions (e.g., DSA, ML, IoT, web development) and behavioral and HR rounds (e.g., STAR-method scenarios, communication, situational judgment).
- **Adaptive Learning Path** – Track performance across sessions and adjust question difficulty and topics to target the candidate's weak areas.

Multi-Agent System

- **Resume Analyzer Agent** – Parses the resume or job title and extracts skills, projects, experience level, and gaps.
- **Interview Knowledge Agent** – Retrieves role-specific questions, industry expectations, and HR guidelines using RAG.
- **Mock Interviewer Agent** – Conducts the interview, asks follow-up questions, and adapts based on the candidate's answers.
- **Evaluation & Feedback Agent** – Scores answers against a rubric, provides model answers, and suggests improvements.
- **Progress Coach Agent** – Builds a personalized preparation plan and tracks improvement over time.

Visualization Dashboard

Show session scores, skill-wise performance (technical, communication, problem-solving, confidence), weak areas, and readiness progress across multiple mock interviews.

Technology

Use of IBM Cloud Lite services / IBM Granite is mandatory (e.g., IBM Granite models on watsonx.ai for generation and evaluation, IBM Cloud Object Storage for question banks and resumes, and a vector database for RAG retrieval).